TEXORA · LECTURE SHEET

Pumps

Machines that pay the friction bill: centrifugal and positive-displacement families, priming, the head–capacity characteristic and the duty point, the affinity laws, specific speed for selection — and cavitation, the suction-side failure that eats impellers, ruled by NPSH.

ME-3203 · THERMAL ENGINEERING AND FLUID MECHANICS · TOPIC 4 · LEVEL 3 TERM 2

1. The two families

Pumps divide by how they add energy. **Rotodynamic** machines — overwhelmingly the **centrifugal pump** — spin the liquid in an impeller and convert velocity to pressure in a volute casing: flow is continuous, output varies with system resistance, and the machine is compact, cheap and tolerant of dirty liquor. **Positive-displacement** machines trap fixed volumes and push them out: reciprocating piston/diaphragm pumps (near-constant flow regardless of pressure — metering and dosing, e.g. chemical dosing in a dye kitchen) and rotary gear/lobe/screw pumps for viscous fluids (printing pastes, resin, softener). Choose by the job: bulk water and liquor circulation is centrifugal territory; precision dosing and thick fluids belong to displacement machines, which must never be throttled on the delivery side — a blocked discharge simply raises pressure until something yields.

A centrifugal pump cannot pump air: it must be **primed** — casing and suction filled with liquid — before it will draw. Flooded suction (liquid supplied under gravity from an overhead tank) is the plant designer’s gift to reliability; self-priming designs exist but flooded suction wins every argument.

2. Head, power and the duty point

A pump is rated by the **total head** it adds, H (metres of liquid), against flow Q. The shaft power to deliver Q against H at efficiency η is:

$$P = \rho g Q H / \eta$$

WORKED EXAMPLE — PUMP SHAFT POWER

A circulation pump delivers Q = 0.05 m³/s (50 L/s) of water against H = 30 m at η = 0.75.
Hydraulic power = ρgQH = 1000 × 9.81 × 0.05 × 30 = 14.7 kW; shaft power = 14.7/0.75 = **19.6**

kW — so a 22 kW motor (with margin) is specified. Note the efficiency's leverage: a worn pump at η = 0.60 would demand 24.5 kW for the same duty, a silent 5 kW tax on every running hour.

A pump does not choose its duty: the **pump characteristic** (H falling as Q rises) intersects the **system characteristic** (static lift + friction ∝ Q², Topic 3) at the **duty point** — the only place the two agree. Throttling a valve steepens the system curve and walks the duty point left (less flow, more head, wasted energy); a variable-speed drive moves the pump curve itself and is the honest way to control flow.

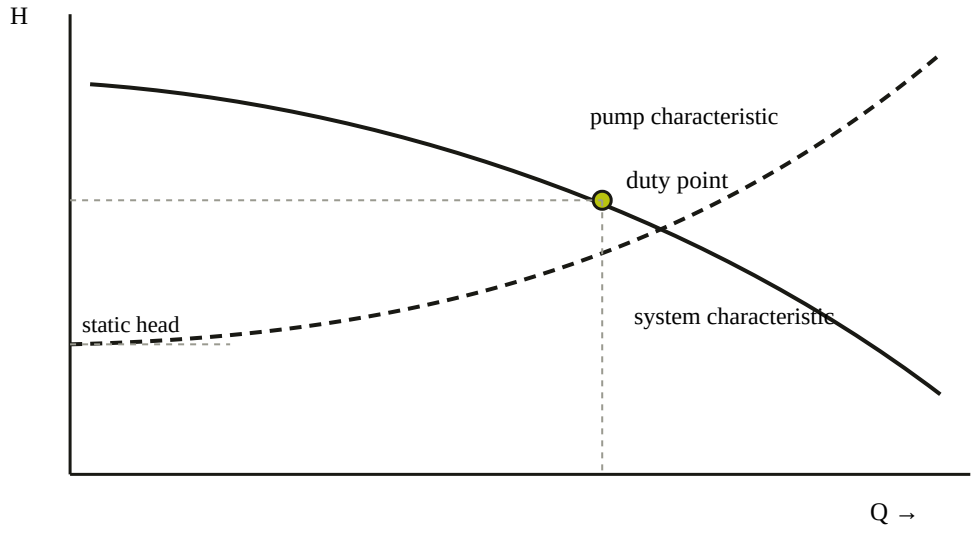


Figure 1 — Pump and system characteristics. The duty point is their intersection: change either curve (speed, valve, pipework) and the duty moves.

3. The affinity laws — what speed changes do

For a given pump at varying speed N (or geometrically similar machines), the homologous quantities scale as:

$$Q \propto N, \quad H \propto N^2, \quad P \propto N^3$$

WORKED EXAMPLE — SPEEDING A PUMP UP

A pump at 1450 rpm delivers 40 L/s against 25 m, absorbing 13 kW. Fitted with a drive at 1750 rpm (ratio 1.207): $Q_2 = 40 \times 1.207 =$ **48.3 L/s**; $H_2 = 25 \times 1.207^2 =$ **36.4 m**; $P_2 = 13 \times 1.207^3 =$ **22.9 kW**. The cube law is the warning label: 21% more speed costs 76% more power — and the mirror image is the energy industry's favourite fact, that slowing a pump 20% nearly halves its power draw.

4. Cavitation and NPSH

If suction pressure falls to the liquid's vapour pressure, vapour bubbles form at the impeller eye and collapse violently downstream — noise like gravel in the casing, vibration, and impeller metal eaten into sponge. This is **cavitation**. Defence is a head budget on the suction side, the **net positive suction head available**:

$$\text{NPSHA} = (p_{atm} - p_{vapour})/\rho g - h_{suction\ lift} - h_{friction}$$

WORKED EXAMPLE — WHY HOT LIQUOR CAVITATES

A pump draws water with 3 m suction lift and 0.8 m of suction friction; the maker requires NPSHR = 4.5 m. At 30 °C (vapour pressure head 0.43 m; atmospheric head 10.3 m): NPSHA = 10.3 − 0.43 − 3 − 0.8 = **6.07 m** — safe, margin 1.57 m. The same pump on 80 °C dye liquor (vapour pressure head 4.83 m): NPSHA = 10.3 − 4.83 − 3 − 0.8 = **1.67 m < 4.5 m — cavitation guaranteed.** Fixes, in order of preference: lower the pump or raise the tank (flooded suction), enlarge/shorten the suction line, cool the liquor, or choose a low-NPSHR pump. Hot-liquor circulation pumps fail in summer for exactly this arithmetic.

Selection closes with **specific speed** $N_s = N\sqrt{Q}/H^{(3/4)}$, which sorts impeller geometry — low values for high-head narrow radial impellers, high values for axial-flow propellers — and confirms the chosen machine is efficient near its duty, not merely capable of it.

5. Key points

- Centrifugal for bulk flow, positive displacement for dosing and viscous duty; displacement pumps must never be dead-headed.
- $P = \rho g Q H/\eta$ — the example duty (50 L/s, 30 m, η 0.75) needs 19.6 kW shaft power.
- The duty point is where pump and system curves cross; throttling wastes energy, speed control moves the pump curve.
- Affinity laws: $Q \propto N$, $H \propto N^2$, $P \propto N^3$ — 1450 → 1750 rpm turns 13 kW into 22.9 kW.
- Cavitation: keep NPSHA > NPSHR with margin; hot liquor is the trap (NPSHA collapses from 6.07 m to 1.67 m between 30 °C and 80 °C).

6. Practice

Answer in your own words; brief model answers follow.

1. Find the shaft power for Q = 0.02 m³/s, H = 45 m, η = 0.70, and the motor size you would specify.
2. A pump runs at 2900 rpm giving 20 L/s at 60 m. Using the affinity laws, state Q and H at 1450 rpm (same system valve fully open — report the homologous point).

3. A pump’s duty point is left of where production wants it. List three physically distinct ways to move it right, and one way that only appears to.
4. Compute NPSHA for a pump drawing 60 °C water (vapour head 2.0 m) with 2 m lift and 0.5 m friction. NPSHR is 5 m. Verdict and two remedies.
5. Why is a centrifugal pump started against a closed or partly closed delivery valve, while a reciprocating pump must never be?
6. A dosing pump must deliver exactly 2 L/min of concentrated alkali against varying back-pressure. Which family, and why does back-pressure not change its delivery?

Answer key

1. Hydraulic = 1000 × 9.81 × 0.02 × 45 = 8.83 kW; shaft = 8.83/0.70 = 12.6 kW. Specify the next standard motor with margin — 15 kW.
2. Speed halves: Q = 20 × 0.5 = 10 L/s; H = 60 × 0.25 = 15 m (the homologous point on the same pump curve family — the actual duty then depends on the system curve).
3. Raise pump speed (VFD); fit a larger impeller diameter / trim in reverse; reduce system resistance (clean strainers, larger or shorter suction/discharge lines, fully open valves) — or put a second pump in parallel for flow. Merely opening hopes: swapping to a bigger motor changes nothing — the pump curve, not the motor, sets the ceiling.
4. NPSHA = 10.3 − 2.0 − 2 − 0.5 = 5.8 m > 5 m: safe, but the margin is only 0.8 m; if margin erodes (strainer fouling), remedies: flood the suction (raise tank/lower pump) or enlarge the suction pipe to cut the 0.5 m friction.
5. The centrifugal pump delivers minimal flow (and minimal power) at closed valve, and cannot build dangerous pressure — its head is capped by the curve. A displacement pump delivers its swept volume regardless: against a closed valve pressure climbs until a relief valve lifts or the pipe, seals or drive fail.
6. A positive-displacement (diaphragm/piston metering) pump: each stroke sweeps a fixed volume and the check valves enforce one-way passage, so delivery is set by stroke rate and volume, essentially independent of discharge pressure (within rating) — exactly what dosing demands.

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